

Assignment – 7

Q1. Write a function that returns the sum of digits of a number using recursion.

```
def sum_of_digits(n):
    if not n:
        return n

    return n % 10 + sum_of_digits(n // 10)

n = int(input("Enter a number: "))
print("Sum of the digits of", n, "are:", sum_of_digits(n))
```

Output:

```
Enter a number: 2503
Sum of the digits of 2503 are: 10
```

Q2. Write a function that checks whether a number is an Armstrong number (without using built in power function).

```
from math import log10

def is_armstrong(n):
    count, res, temp = int(log10(n) + 1), 0, n

    while temp:
        digit, power = temp % 10, 1

        for i in range(count):
            power *= digit

        res += power
        temp //= 10

    return n == res

n = int(input("Enter a number: "))
print("Is", n, "an armstrong number?", is_armstrong(n))
```

Output:

```
Enter a number: 2503
Is 2503 an armstrong number? False
```

Q3. Write a function that returns the largest digit in a number (without converting to string).

```
def largest_digit(n):
    res = 0
```

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

```

while n:
    res = max(res, n % 10)
    n //= 10

return res

n = int(input("Enter a number: "))
print("Largest digit in", n, "is:", largest_digit(n))

```

Output:

Enter a number: 2503
 Largest digit in 2503 is: 5

Q4. Write a function that takes a number and returns True if the number is palindrome and False otherwise (without using string conversion).

```

def is_palindrome(n):
    rev, temp = 0, n

    while temp:
        rev = rev * 10 + temp % 10
        temp //= 10

    return rev == n

n = int(input("Enter a number: "))
print("Is", n, "a palindrome number?", is_palindrome(n))

```

Output:

Enter a number: 2503
 Is 2503 a palindrome number? False

Q5. Write a function that returns the count of prime digits in a number.

```

def prime_count(n):
    res = 0

    while n:
        res += n % 10 in (2, 3, 5, 7)
        n //= 10

    return res

n = int(input("Enter a number: "))
print("No. of prime digits in", n, "are:", prime_count(n))

```

Output:

Enter a number: 2503
 No. of prime digits in 2503 are: 3

Name: Asmit Ayush
 SIC No: 24BCSE59
 Lab Roll No: 21

Q6. Write a function that rotates digits of a number k times to the right.

Example: rotate(12345,2) → 45123

```
def right_rotate(n, k):
    n = str(n)
    k %= len(n)
    return int(n[-k:] + n[:-k])

n = int(input("Enter a number: "))
k = int(input("Enter no. of places to rotate: "))
print("Right rotation of", n, "by", k, "places is:", right_rotate(n, k))
```

Output:

```
Enter a number: 2503
Enter no. of places to rotate: 2
Right rotation of 2503 by 2 places is: 325
```

Q7. Write a function that returns the factorial of a number using recursion without using loops.

```
def factorial(n):
    if n <= 1:
        return 1

    return n * factorial(n - 1)

n = int(input("Enter a number: "))
print("Factorial of", n, "is:", factorial(n))
```

Output:

```
Enter a number: 5
Factorial of 5 is: 120
```

Q8. Write a function that checks whether a number is a Perfect Number.

```
def is_perfect(n):
    res = 1

    for i in range(2, n // 2 + 1):
        if n % i == 0:
            res += i

    return res == n

n = int(input("Enter a number: "))
print("Is", n, "an prefect number?", is_perfect(n))
```

Output:

```
Enter a number: 2503
Is 2503 an prefect number? False
```

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

Q9. Write a function that returns the GCD of two numbers using recursion.

```
def gcd(a, b):
    if b == 0:
        return a

    return gcd(b, a % b)

a = int(input("Enter the first number: "))
b = int(input("Enter the second number: "))
print("GCD of", a, "and", b, "is:", gcd(a, b))
```

Output:

```
Enter the first number: 6
Enter the second number: 7
GCD of 6 and 7 is: 1
```

Q10. Write a function that returns the LCM of two numbers using the GCD function.

```
def gcd(a, b):
    if not b:
        return a

    return gcd(b, a % b)

lcm = lambda a, b: a * b // gcd(a, b)
a = int(input("Enter the first number: "))
b = int(input("Enter the second number: "))
print("LCM of", a, "and", b, "is:", lcm(a, b))
```

Output:

```
Enter the first number: 6
Enter the second number: 7
LCM of 6 and 7 is: 42
```

Q11. Write a function that returns the second largest element in a list without using `sort()` or `max()`.

```
def second_largest(lst):
    res = largest = lst[0]

    for num in lst[1:]:
        if num > largest:
            res = largest
            largest = num

        elif num > res and num != largest:
            res = num

    return res
```

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

```
lst = list(eval(input("Enter a list of numbers: ")))
print("Second largest number in", lst, "is:", second_largest(lst))
```

Output:

Enter a list of numbers: 1, 2, 3, 4, 5
 Second largest number in [1, 2, 3, 4, 5] is: 4

Q12. Write a function that counts the number of vowels in a string without using built-in count().

```
def vowel_count(string):
    res = 0

    for ch in string.lower():
        res += ch in ('a', 'e', 'i', 'o', 'u')

    return res

string = input("Enter a string: ")
print("No. of vowels in", string, "are:", vowel_count(string))
```

Output:

Enter a string: abcdefghi
 No. of vowels in abcdefghi are: 3

Q13. Write a function that removes duplicate elements from a list without using `set()`.

```
def remove_duplicate(lst):
    res = []

    for num in lst:
        if num not in res:
            res.append(num)

    return res

lst = list(eval(input("Enter a list of numbers: ")))
print("After removing duplicate elements from", lst, "is:", remove_duplicate(lst))
```

Output:

Enter a list of numbers: 1, 5, 9, 7, 5, 3, 2, 1
 After removing duplicate elements from [1, 5, 9, 7, 5, 3, 2, 1] is: [1, 5, 9, 7, 3, 2]

Q14. Write a function that returns the frequency of each character in a string.

```
def character_frequency(string):
    return {ch: string.count(ch) for ch in string}

string = input("Enter a string: ")
print("Frequency of every character in", string, "are:")
print(character_frequency(string))
```

Output:

```
Enter a string: swapnaraj
Frequency of every character in swapnaraj are:
{'s': 1, 'w': 1, 'a': 3, 'p': 1, 'n': 1, 'r': 1, 'j': 1}
```

Q15. Write a function that checks whether two strings are anagrams (without using sorted()).

```
def is_anagram(string1, string2):
    lst1, lst2 = list(string1), list(string2)
    lst1.sort()
    lst2.sort()
    return lst1 == lst2

string1 = input("Enter the first string: ")
string2 = input("Enter the second string: ")
print("Is", string1, "and", string2, "are anagrams?", is_anagram(string1, string2))
```

Output:

```
Enter the first string: dreamc
Enter the second string: creamd
Is dreamc and creamd are anagrams? True
```

Q16. Write a function that returns the sum of factorial of each digit of a number.

```
from math import factorial

def factorial_sum(n):
    res = 0

    while n:
        res += factorial(n % 10)
        n //= 10

    return res

n = int(input("Enter a number: "))
print("Factorial of every digit of", n, "is:", factorial_sum(n))
```

Output:

```
Enter a number: 2503
Factorial of every digit of 2503 is: 129
```

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

Q17. Write a function that returns the nth Fibonacci number using recursion.

```
def nth_fibonacci(n):
    if n <= 1:
        return n

    return nth_fibonacci(n - 1) + nth_fibonacci(n - 2)

n = int(input("Enter a number: "))
print(n, "th fibonacci number is: ", nth_fibonacci(n), sep="")
```

Output:

Enter a number: 7
7th fibonacci number is: 13

Q18. Write a function that returns the next greater number formed using the same digits

Input: 5342

Output: 5423.

```
def next_greatest(n):
    res = n + 1
    sort = str(sorted(str(n)))

    if str(n) == sort:
        return n

    while str(sorted(str(res))) != sort:
        res += 1

    return res

n = int(input("Enter a number: "))
print("Next greatest number after", n, "with same digits is:", next_greatest(n))
```

Output:

Enter a number: 2503
Next greatest number after 2503 with same digits is: 2530

Q19. Write a lambda function to return the square of a number.

```
squared = lambda x: x ** 2
n = int(input("Enter a number: "))
print(n, "squared is:", squared(n))
```

Output:

Enter a number: 2503
2503 squared is: 6265009

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

Q20. Write a lambda function to find the maximum of three numbers.

```
maximum = lambda a, b, c: max(a, b, c)
a = int(input("Enter the first number: "))
b = int(input("Enter the second number: "))
c = int(input("Enter the third number: "))
print("Maximum of", a, b, c, "is:", maximum(a, b, c))
```

Output:

```
Enter the first number: 2
Enter the second number: 3
Enter the third number: 5
Maximum of 2 3 5 is: 5
```

Q21. Write a lambda function to check whether a number is even or odd.

```
is_even = lambda x: x % 2 == 0
n = int(input("Enter a number: "))
print("Is", n, "an even number?", is_even(n))
```

Output:

```
Enter a number: 2503
Is 2503 an even number? False
```

Q22. Write a lambda function to reverse a string.

```
reverse = lambda string: string[::-1]
string = input("Enter a string: ")
print("Reverse of", string, "is:", reverse(string))
```

Output:

```
Enter a string: dream
Reverse of dream is: maerd
```

Q23. Write a lambda function that returns True if a number is divisible by both 3 and 5.

```
is_divisible = lambda x: x % 15 == 0
n = int(input("Enter a number: "))
print("Is", n, "divisible by 3 and 5?", is_divisible(n))
```

Output:

```
Enter a number: 2503
Is 2503 divisible by 3 and 5? False
```

Q24. Given a list of integers, use map() to create a new list containing the cube of each number.

```
lst = list(eval(input("Enter a list of numbers: ")))
print("List of numbers cubed are:", list(map(lambda x: x ** 3, lst)))
```

Output:

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

Enter a list of numbers: 1, 2, 3, 4, 5

List of numbers cubed are: [1, 8, 27, 64, 125]

Q25. Given a list of strings, use map() to convert all strings into uppercase.

```
lst = list(eval(input("Enter a list of strings: ")))
print("List of string to uppercase are:", list(map(str.upper, lst)))
```

Output:

Enter a list of strings: "dream", "hello"

List of string to uppercase are: ['DREAM', 'HELLO']

Q26. Convert a list of temperatures in Celsius to Fahrenheit using map().

```
lst = list(eval(input("Enter a list of temperature in Celsius: ")))
print("List of temperature in Fahrenheit are:", list(map(lambda x: x * 9 / 5 + 32, lst)))
```

Output:

Enter a list of temperature in Celsius: 1, 2, 3, 4, 5

List of temperature in Fahrenheit are: [33.8, 35.6, 37.4, 39.2, 41.0]

Q27. Given a list of numbers, add 10 to every element using map() and lambda.

```
lst = list(eval(input("Enter a list of numbers: ")))
print("List of numbers with 10 added are:", list(map(lambda x: x + 10, lst)))
```

Output:

Enter a list of numbers: 1, 2, 3, 4, 5

List of numbers with 10 added are: [11, 12, 13, 14, 15]

Q28. Given two lists:

L1 = [1,2,3,4]

L2 = [5,6,7,8]

Use map() to create a new list containing sum of corresponding elements.

```
lst1 = list(eval(input("Enter the first list of numbers: ")))
lst2 = list(eval(input("Enter the second list of numbers: ")))
lst = list(map(lambda x: x[0] + x[1], zip(lst1, lst2)))
print("List of numbers with corresponding elements added are:", lst)
```

Output:

Enter the first list of numbers: 1, 2, 3, 4, 5

Enter the second list of numbers: 6, 7, 8, 9, 10

List of numbers with corresponding elements added are: [7, 9, 11, 13, 15]

Q29. From a list of numbers, use filter() to extract all even numbers.

```
lst = list(eval(input("Enter a list of numbers: ")))
print("List of even numbers are:", list(filter(lambda x: x % 2 == 0, lst)))
```

Output:

Enter a list of numbers: 1, 2, 3, 4, 5
List of even numbers are: [2, 4]

Q30. From a list, use filter() to remove all negative numbers.

```
lst = list(eval(input("Enter a list of numbers: ")))
print("List of non-negative numbers are:", list(filter(lambda x: x >= 0, lst)))
```

Output:

Enter a list of numbers: 1, 2, 3, 4, 5
List of non-negative numbers are: [1, 2, 3, 4, 5]

Q31. Given a list of integers, use filter() to extract all prime numbers.

```
def is_prime(num):
    for i in range(2, num // 2 + 1):
        if num % i == 0:
            return False

    return True

lst = list(eval(input("Enter a list of numbers: ")))
print("List of prime numbers are:", list(filter(is_prime, lst)))
```

Output:

Enter a list of numbers: 1, 2, 3, 4, 5
List of prime numbers are: [1, 2, 3, 5]

Q32. Given a string, use filter() to keep only vowels.

```
string = input("Enter a string: ")
string = ''.join(filter(lambda ch: ch.lower() in ('a', 'e', 'i', 'o', 'u'), string))
print("String with only vowel:", string)
```

Output:

Enter a string: dream
String with only vowel: ea

Q33. From a list of words, use filter() to keep words having length greater than 5.

```
lst = list(eval(input("Enter a list of words: ")))
print("List of words with length greater than 5 are:", list(filter(lambda x: len(x) > 5, lst)))
```

Output:

Enter a list of words: "hello", "world", "dream"

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

List of words with length greater than 5 are: []

Q34. Use reduce() to find the sum of all elements in a list.

```
from functools import reduce
```

```
lst = list(eval(input("Enter a list of numbers: ")))  
print("Sum of all the elements are:", reduce(lambda x, y: x + y, lst))
```

Output:

```
Enter a list of numbers: 1, 2, 3, 4, 5  
Sum of all the elements are: 15
```

Q35. Use reduce() to find the product of all numbers in a list.

```
from functools import reduce
```

```
lst = list(eval(input("Enter a list of numbers: ")))  
print("Product of all the elements are:", reduce(lambda x, y: x * y, lst))
```

Output:

```
Enter a list of numbers: 1, 2, 3, 4, 5  
Product of all the elements are: 120
```

Q36. Use reduce() to find the largest element in a list.

```
from functools import reduce
```

```
lst = list(eval(input("Enter a list of numbers: ")))  
print("Largest elements is:", reduce(lambda x, y: max(x, y), lst))
```

Output:

```
Enter a list of numbers: 1, 2, 3, 4, 5  
Largest elements is: 5
```

Q37. Given a list of digits [1,2,3,4], use reduce() to form the number 1234.

```
from functools import reduce
```

```
lst = list(eval(input("Enter a list of numbers: ")))  
print("Result:", reduce(lambda x, y: x * 10 + y, lst))
```

Output:

```
Enter a list of numbers: 1, 2, 3, 4, 5  
Result: 12345
```

Q38. Use reduce() to count the total number of characters in a list of strings.

Name: Asmit Ayush
SIC No: 24BCSE59
Lab Roll No: 21

```
from functools import reduce
```

```
string = input("Enter a string: ")
print("Result:", reduce(lambda x, y: x + 1, lst, 0))
```

Output:

```
Enter a string: dream
Result: 5
```

Q39. From a list, use filter() to select numbers divisible by 3, then map() to square them.

```
lst = list(eval(input("Enter a list of numbers: ")))
print("Resultant list:", list(map(lambda x: x ** 2, filter(lambda x: x % 3 == 0, lst))))
```

Output:

```
Enter a list of numbers: 1, 2, 3, 4, 5
Resultant list: [9]
```

Q40. Given a list of integers, use map() to double each number and reduce() to find their sum.

```
from functools import reduce

lst = list(eval(input("Enter a list of numbers: ")))
print("Result:", reduce(lambda x, y: x + y, map(lambda x: x * 2, lst)))
```

Output:

```
Enter a list of numbers: 1, 2, 3, 4, 5
Result: 30
```

Q41. From a list of words, keep only palindromes using filter() and lambda.

```
lst = list(eval(input("Enter a list of words: ")))
print("List of words that are palindrome are:", list(filter(lambda x: x == x[::-1], lst)))
```

Output:

```
Enter a list of words: "hello", "world", "dream"
List of words that are palindrome are: []
```

Q42. Given a sentence, use map() to find the length of each word and reduce() to get the total characters.

```
sentence = input("Enter a string: ")
print("Total length:", reduce(lambda x, y: x + y, map(len, sentence.split())))
```

Output:

```
Enter a string: dream
Total length: 5
```

Q43. From a list of numbers:

- remove odd numbers (filter)
- cube the remaining numbers (map)
- find their sum (reduce)

```
from functools import reduce
```

```
lst = list(eval(input("Enter a list of numbers: ")))  
res = reduce(lambda x, y: x + y, map(lambda x: x ** 3, filter(lambda x: x % 2 == 0, lst)))  
print("Result:", res)
```

Output:

Enter a list of numbers: 1, 2, 3, 4, 5

Result: 72